INTRODUCTION

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today's digital era, cyber threats are increasing rapidly due to the widespread use of the internet, cloud computing, and connected devices. Organizations, businesses, and individuals rely heavily on digital systems to store sensitive data, making them vulnerable to cyber-attacks such as malware, phishing, ransomware, and unauthorized access. Traditional security systems often struggle to detect new and evolving threats because they mainly rely on predefined rules and signatures. This creates a need for intelligent and adaptive security solutions.

The AI-Based Cyber Security Threat Detection System is designed to address these challenges by using Artificial Intelligence and Machine Learning techniques to identify and prevent cyber threats in real-time. This system analyzes network traffic, user behavior, and system activities to detect suspicious patterns and potential security breaches. By leveraging AI algorithms, the system can learn from past data, improve accuracy, and detect unknown threats more effectively than traditional security methods.

The main objective of this project is to develop a smart and automated cybersecurity solution that enhances system security, reduces human intervention, and improves threat detection efficiency. The proposed system will monitor activities continuously, detect anomalies, generate alerts, and help administrators take appropriate actions to prevent attacks.

This project aims to improve cybersecurity infrastructure by implementing intelligent threat detection mechanisms, ensuring data protection, and maintaining system integrity. It also provides scalability and adaptability, making it suitable for organizations of different sizes.

Objective

The AI-Based Cyber Security Threat Detection System aims to provide an intelligent and automated security solution for detecting cyber threats in real-time. Traditional security systems rely on predefined rules, which makes them less effective against modern and evolving cyber threats. Managing cybersecurity manually is time-consuming, error-prone, and inefficient. With the help of Artificial Intelligence and Machine Learning, threats can be detected faster, more accurately, and automatically.

The objectives of this project are mentioned below:

• To develop an AI-based system for detecting cyber security threats
• To monitor user activities and network traffic in real-time
• To detect suspicious login attempts and unauthorized access
• To provide OTP-based secure authentication system
• To create an admin panel for monitoring user activity
• To detect multiple failed login attempts and lock user accounts
• To generate alerts for suspicious behavior
• To maintain logs of user login history and session duration
• To improve system security using intelligent threat detection

Proposed System

The proposed AI-Based Cyber Security Threat Detection System is designed to provide a smart and secure platform that monitors system activities and detects threats automatically. The system includes a secure login and signup system with OTP verification through email and mobile.

The system will include an admin panel to monitor users, track login attempts, and detect suspicious activities. If a user attempts to log in more than three times with incorrect credentials, the account will be automatically locked, and the admin will be notified.

The system will store user data securely in a database and provide secure authentication. The AI-based detection system will analyze login patterns and identify abnormal behavior.

The system will help organizations and users improve security and prevent cyber-attacks.

Purpose

The purpose of this project is to develop an AI-based cybersecurity system that detects threats and protects sensitive data. The system allows administrators to monitor user activity, detect suspicious login attempts, and manage security alerts.

The system also provides:

- Secure login and signup system
- OTP verification (Email and Mobile)
- Admin dashboard for monitoring users
- Failed login detection system
- Account lock after multiple failed attempts
- Login history tracking
- Real-time threat detection
- Secure authentication system
- User session monitoring

Scope

The AI-Based Cyber Security Threat Detection System can be accessed using secure login credentials. Only authorized administrators can manage the system and unlock user accounts. Users can securely log in and access the system.

The scope of the system includes:

- User Registration System
- Secure Login System
- OTP Verification (Email & Mobile)
- Admin Panel
- User Activity Monitoring
- Failed Login Detection
- Account Lock System
- Login History Management
- Threat Detection System
- Secure Database Management
- Real-Time Alerts
- Responsive Web Design

PRELIMINARY SYSTEM ANALYSIS

Preliminary Investigation

The preliminary investigation of the AI-Based Cyber Security Threat Detection System shows that many organizations still rely on traditional security mechanisms such as firewalls, antivirus software, and manual monitoring to detect cyber threats. These traditional systems are limited in detecting modern and advanced cyber-attacks. As cyber threats continue to evolve, traditional security systems become less effective and fail to detect unknown or sophisticated attacks.

Manual monitoring of system logs and user activities is time-consuming and prone to human error. Security administrators often struggle to identify suspicious behavior in real-time, which increases the risk of security breaches. Therefore, there is a need for an intelligent and automated cybersecurity system that can detect threats efficiently and in real-time.

Present System in Use

In the present system, organizations use traditional cybersecurity solutions such as antivirus software, firewalls, and manual monitoring systems. These systems rely mainly on predefined rules and signature-based detection techniques. While these methods are effective against known threats, they often fail to detect new and unknown cyber-attacks.

Additionally, most existing systems do not provide advanced monitoring of user behavior, login attempts, or suspicious activities. There is limited automation, and administrators must manually review logs and alerts, which makes the process inefficient and time-consuming.

Flaws in Present System

The existing system has several limitations, including:

- Lack of real-time threat detection
- Inability to detect unknown cyber threats
- Manual monitoring of system activities
- Limited automation and intelligence
- No user behavior analysis
- No login attempt monitoring
- Lack of account lock system

- Delayed threat response
- Increased risk of security breaches

Need of New System

To overcome the limitations of the existing system, there is a need for an AI-Based Cyber Security Threat Detection System. The proposed system will use Artificial Intelligence and Machine Learning techniques to detect cyber threats automatically and in real-time.

The system will provide advanced features such as:

- AI-based threat detection
- Real-time monitoring
- Suspicious activity detection
- Secure login and authentication
- Account lock after multiple failed attempts
- Admin panel for monitoring users
- Alert and notification system
- User behavior analysis

Feasibility Study

The proposed system is feasible in the following ways:

Technical Feasibility

The system can be developed using technologies such as:

- HTML
- CSS
- JavaScript
- Python
- MySQL Database
- XAMPP Server
- Machine Learning Libraries (Scikit-learn, TensorFlow)

Economic Feasibility

The system is economically feasible because it uses open-source tools and technologies. The development cost is minimal, and maintenance cost is low. No additional hardware requirements are needed.

Operational Feasibility

The system is operationally feasible as it is user-friendly and easy to operate. The admin panel allows administrators to monitor activities easily, and users can access the system securely.

Project Category

The AI-Based Cyber Security Threat Detection System is categorized as:

- Web-Based Application
- Cyber Security System
- AI-Based Detection System
- Network Security Application
- Threat Detection and Monitoring System

SOFTWARE ENGINEERING PARADIGAM APPLIED

SOFTWARE ENGINEERING PARADIGM APPLIED

The Software Engineering Paradigm applied to the AI-Based Cyber Security Threat Detection System is the Waterfall Model. The Waterfall Model is a structured and sequential approach where development is carried out in different phases such as requirement analysis, system design, implementation, testing, deployment, and maintenance. This model is suitable for this project because the requirements of the cybersecurity system are clearly defined, and development follows a systematic process.

In the first phase, Requirement Analysis, system requirements were gathered based on the need for a secure login system, AI-based threat detection, user activity monitoring, and admin panel management. The functionalities and system requirements were identified and documented properly.

In the System Design Phase, the architecture of the system was designed, including modules such as login system, threat detection module, admin panel, and database management system.

During the Implementation Phase, the system was developed using technologies such as HTML, CSS, JavaScript, Python, MySQL, and XAMPP server.

In the Testing Phase, the system was tested for bugs, errors, and security vulnerabilities. After testing, the system was deployed and maintained for future updates.

MODULES

The AI-Based Cyber Security Threat Detection System is divided into several modules to ensure smooth functioning and efficient management of the system. Each module performs specific tasks and together they form a complete cybersecurity system.

Admin Module

The Admin Module is responsible for managing the entire system. The administrator can log in securely using username and password. After logging in, the admin can monitor user activities, view login history, detect suspicious activities, and unlock user accounts.

The admin can also:

- View user login details
- Monitor failed login attempts
- Unlock locked user accounts
- View alerts and suspicious activity reports
- Manage users and system security

User Authentication Module

The User Authentication Module provides secure login and signup functionality. Users can register using email and mobile number with OTP verification. This module ensures that only authorized users can access the system.

This module includes:

• User Registration
• Login System
• OTP Verification (Email & Mobile)
• Password Encryption
• Account Lock after multiple failed attempts

Threat Detection Module

The Threat Detection Module is the core module of the system. It uses Artificial Intelligence and Machine Learning algorithms to detect suspicious behavior and cyber threats.

This module performs:

• Suspicious login detection
• Multiple failed login attempt detection
• Abnormal user behavior detection
• Real-time monitoring
• Threat alerts generation

User Activity Monitoring Module

This module monitors user activities such as login time, logout time, session duration, and login attempts. It helps in detecting suspicious activities and unauthorized access.

Features include:

• Login tracking
• Session monitoring
• IP address tracking
• Activity logging
• Suspicious behavior detection

Alert and Notification Module

The Alert and Notification Module generates alerts when suspicious activities are detected. These alerts are sent to the administrator for immediate action.

Features include:

- Suspicious login alert
- Multiple failed attempts alert
- Unauthorized access alert
- System notification

Database Management Module

The Database Module stores all system data such as:

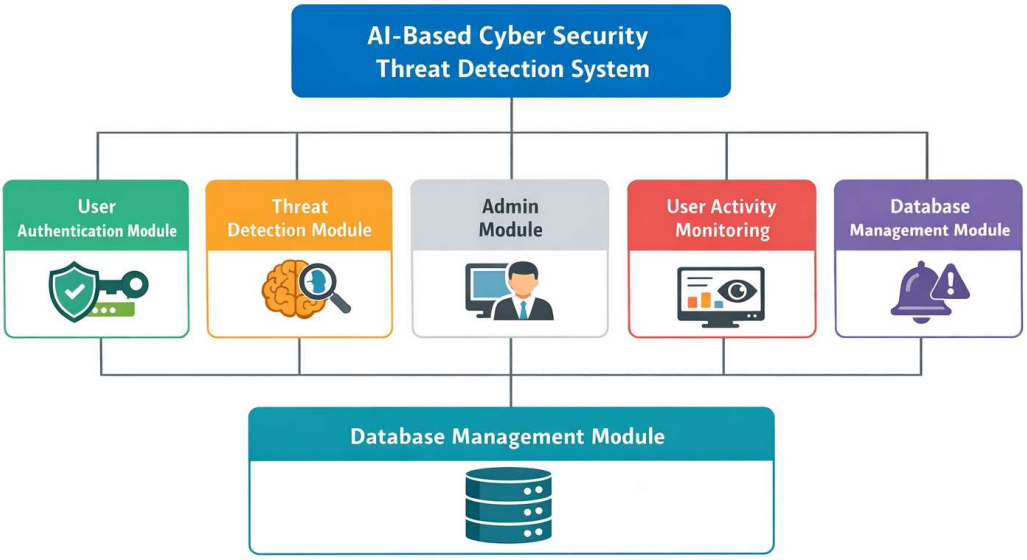
- User information
- Login history
- Activity logs
- Threat detection records
- Admin details

SYSTEM

The AI-Based Cyber Security Threat Detection System is designed as a web-based application that consists of multiple modules working together to provide intelligent security protection. The system mainly includes Admin Panel, User Panel, Authentication Module, Threat Detection Module, and Database Module.

The Admin Panel allows the administrator to monitor user activities, detect threats, and manage system security. The User Panel allows users to register, login, and access the system securely. The Authentication Module provides secure login functionality and OTP verification. The Threat Detection Module identifies suspicious activities and cyber threats using AI techniques. The Database Module stores all user data and system logs securely.

MODULAR CHART



SOFTWARE & HARDWARE REQUIRMENT SPESIFICATON

SOFTWARE & HARDWARE REQUIREMENT SPECIFICATION

Software Requirement Specification

The software requirements specify the software components needed for developing and running the Portfolio Website System. The system is developed using modern web technologies that ensure flexibility, efficiency, and user-friendly interaction.

Operating System

- Windows 10 or Above
- Linux (Optional)

Frontend Technologies

- HTML5
- CSS3
- JavaScript

Backend Technologies

- Python
- Flask Framework

Database

- MySQL Database

Development Tools

- Visual Studio Code
- XAMPP Server
- Web Browser (Google Chrome / Microsoft Edge)

Additional Software Requirements

- Python 3.x
- MySQL Server
- Git (Optional for version control)

Hardware Requirement Specification

The hardware requirements define the minimum system configuration needed to develop and run the Portfolio Website System.

Minimum Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB Minimum
- Hard Disk: 500 GB
- Monitor: 15-inch display
- Keyboard and Mouse

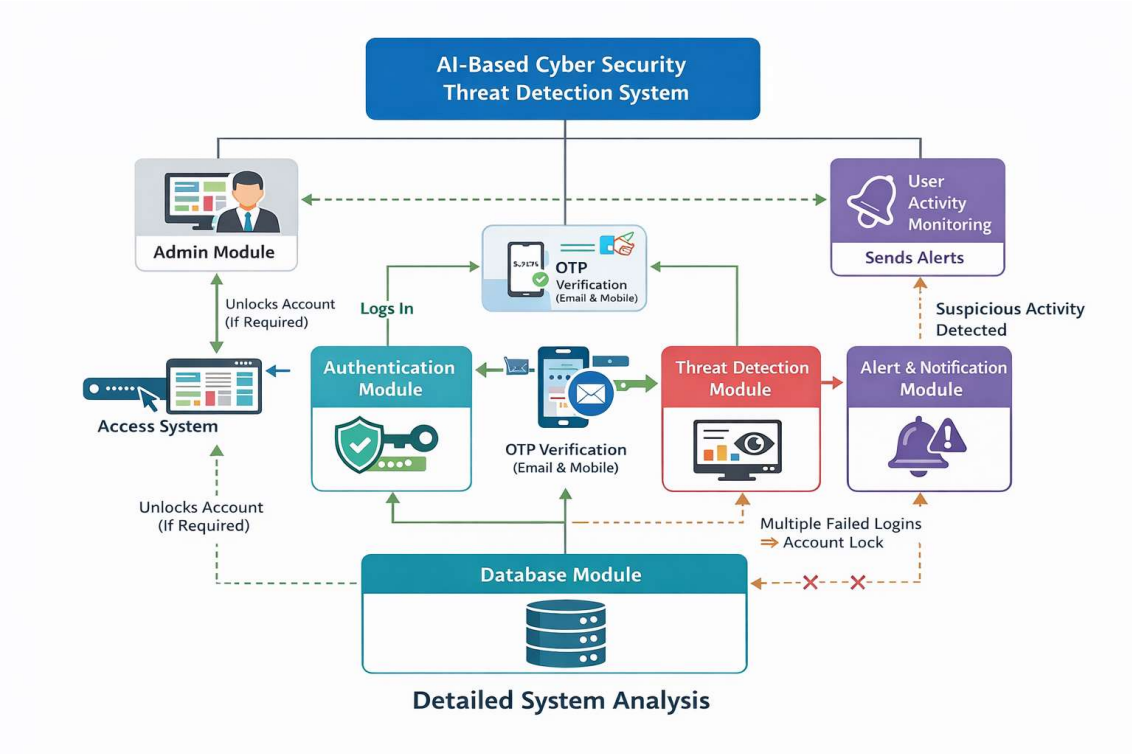
System Requirements Summary

Requirement Type	Specification
Operating System	Windows 10 / Windows 11 / Linux
Frontend	HTML, CSS, JavaScript
Backend	Python (Flask)
Database	MySQL
Server	XAMPP
RAM	4 GB Minimum
Processor	Intel i3 or above
Additional	Machine Learning Libraries

DETAILED SYSTEM ANALYSIS

DETAILED SYSTEM ANALYSIS

Detailed System Analysis is an important phase of system development that focuses on understanding system requirements, user needs, and system functionalities. The AI-Based Cyber Security Threat Detection System is designed to provide intelligent security monitoring and threat detection using Artificial Intelligence techniques. This phase helps in identifying system components, functional requirements, and data flow within the system.



The system analysis ensures that the proposed cybersecurity system is efficient, reliable, and capable of detecting cyber threats in real-time.

Numbers of Modules and Process Logic

The AI-Based Cyber Security Threat Detection System is divided into several modules to ensure smooth and secure functioning. Each module performs specific tasks and interacts with other modules to complete system operations.

Number of Modules

The AI-Based Cyber Security Threat Detection System consists of the following main modules:

1. Admin Module

This module allows the administrator to manage the entire system. The admin can log in securely and monitor user activities, view login attempts, unlock user accounts, and manage security alerts. The admin also has access to the dashboard and system control panel.

2. User Module

This module allows users to register and log into the system securely. Users can access the system after successful authentication. The system monitors user activities and detects suspicious behavior.

3. Authentication Module

This module handles secure login and signup functionality. It includes OTP verification using email and mobile number. It also handles password encryption and account locking after multiple failed login attempts.

4. Threat Detection Module

This module is responsible for detecting suspicious activities and cyber threats using Artificial Intelligence and Machine Learning techniques. It monitors user behavior, login attempts, and abnormal activities.

5. User Activity Monitoring Module

This module monitors user activities such as login time, logout time, session duration, and failed login attempts. This module helps in detecting suspicious behavior.

6. Alert and Notification Module

This module generates alerts when suspicious activities or threats are detected. The admin receives notifications for immediate action.

7. Database Module

This module stores all system data such as user information, login history, activity logs, and threat detection records.

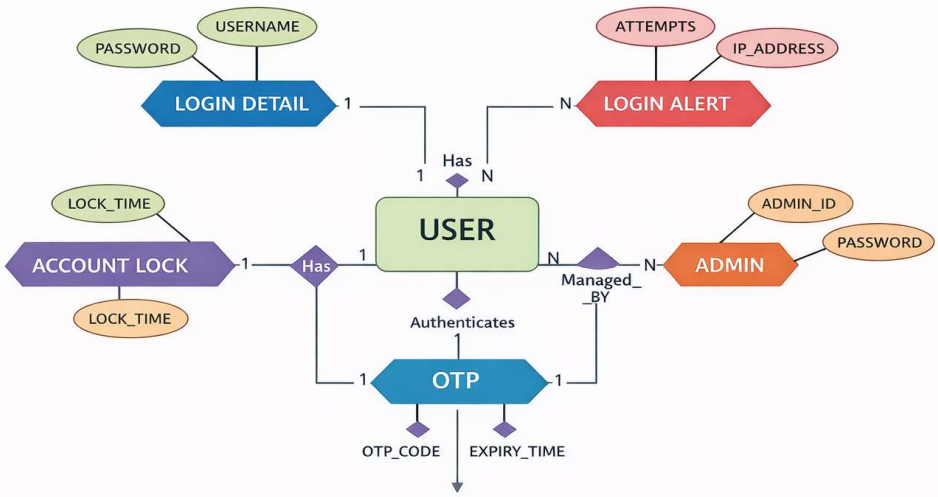
Process Logic

The process logic of the AI-Based Cyber Security Threat Detection System is described below:

1. The user accesses the system through a web browser.
2. The user registers using email and mobile number.
3. OTP verification is performed for authentication.
4. After successful registration, the user logs into the system.
5. The system monitors user activities continuously.
6. If suspicious activity is detected, the system generates an alert.
7. If the user enters wrong credentials more than three times, the account is locked.
8. The admin logs into the admin dashboard.
9. The admin monitors login history and suspicious activities.
10. The admin unlocks user accounts if required.
11. All system data is stored in the database.
12. The system continuously monitors and updates security status.

Data Structures and Tables

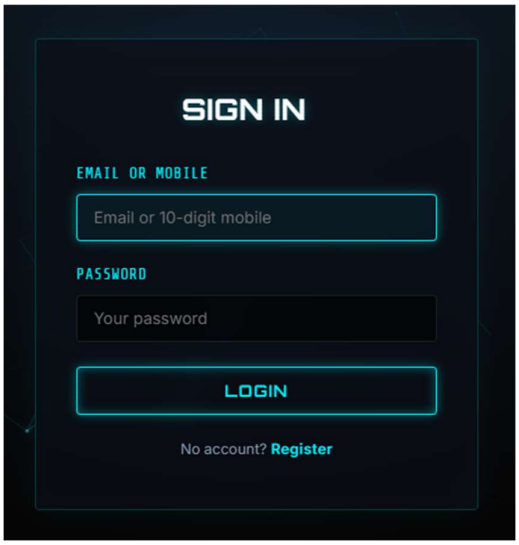
Table	Action	Rows	Type	Collation	Size	Overhead
admin	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	32.0 KiB	-
login_logs	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	32.0 KiB	-
otp_store	Browse Structure Search Insert Empty Drop	14	InnoDB	utf8mb4_general_ci	16.0 KiB	-
support_tickets	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	32.0 KiB	-
users	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	48.0 KiB	-
user_requests	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	32.0 KiB	-
6 tables	**Sum**	**23**	**InnoDB**	**utf8mb4_general_ci**	**192.0 KiB**	**0 B**

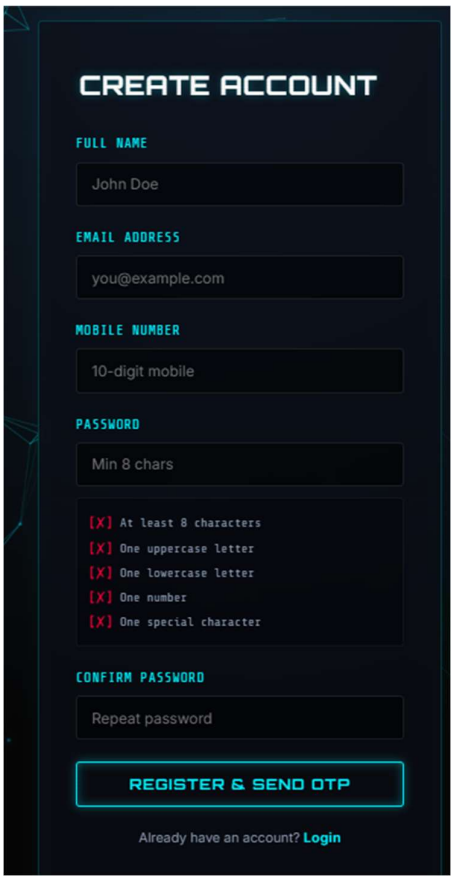


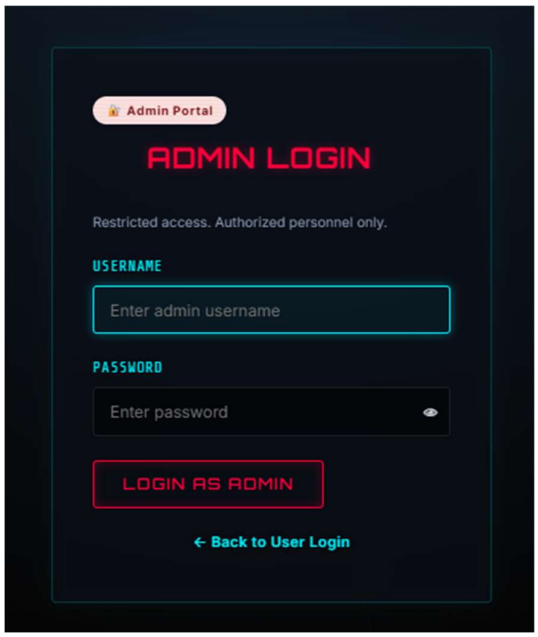
Entity Relationship Diagram

SYSTEM DESIGN

FORM DESIGN


SIGN IN
EMAIL OR MOBILE
Email or 10-digit mobile
PASSWORD
Your password
LOGIN
No account? Register


CREATE ACCOUNT
FULL NAME
John Doe
EMAIL ADDRESS
you@example.com
MOBILE NUMBER
10-digit mobile
PASSWORD
Min 8 chars
[X] At least 8 characters
[X] One uppercase letter
[X] One lowercase letter
[X] One number
[X] One special character
CONFIRM PASSWORD
Repeat password
REGISTER & SEND OTP
Already have an account? Login


Admin Portal
ADMIN LOGIN
Restricted access. Authorized personnel only.
USERNAME
Enter admin username
PASSWORD
Enter password
LOGIN AS ADMIN
← Back to User Login

SOURCE CODE

```
from flask import Flask, render_template, request, redirect, url_for, session, flash
from flask_wtf.csrf import CSRFProtect
import bcrypt, random, string, smtplib
from email.mime.text import MIMEText
from email.mime.multipart import MIMEMultipart
from datetime import datetime, timedelta
from functools import wraps
from db import get_db
from config import Config

app = Flask(__name__)
app.config.from_object(Config)
csrf = CSRFProtect(app)

# ── Helpers ──────────────────────────────────────────────────────────────

def generate_otp(length=6):
    return ''.join(random.choices(string.digits, k=length))

def save_otp(identifier, otp, otp_type):
    db = get_db()
    try:
        with db.cursor() as cur:
            cur.execute("UPDATE otp_store SET used=1 WHERE identifier=%s AND
otp_type=%s AND used=0",
                        (identifier, otp_type))
            expires = datetime.now() + timedelta(minutes=Config.OTP_EXPIRY_MINUTES)
```

```
        cur.execute("INSERT INTO otp_store (identifier, otp, otp_type, expires_at) VALUES (%s,%s,%s,%s)",
                    (identifier, otp, otp_type, expires))
    db.commit()
  finally:
    db.close()


def verify_otp(identifier, otp, otp_type):
  db = get_db()
  row = None
  try:
    with db.cursor() as cur:
      cur.execute("""SELECT id FROM otp_store
                    WHERE identifier=%s AND otp=%s AND otp_type=%s
                     AND used=0 AND expires_at > NOW()
                    ORDER BY id DESC LIMIT 1""",
                   (identifier, otp, otp_type))
      row = cur.fetchone()
      if row:
        cur.execute("UPDATE otp_store SET used=1 WHERE id=%s", (row['id'],))
    db.commit()
  finally:
    db.close()
  return bool(row)


def send_email_otp(email, otp):
  msg = MIMEMultipart()
  msg['From'] = Config.GMAIL_USER
```

```
    msg['To'] = email
    msg['Subject'] = 'Your OTP Code'
    msg.attach(MIMEText(
        f"Your OTP is: {otp}\nValid for {Config.OTP_EXPIRY_MINUTES} minutes.", 'plain'
    ))
    with smtplib.SMTP('smtp.gmail.com', 587) as server:
        server.ehlo()
        server.starttls()
        server.login(Config.GMAIL_USER, Config.GMAIL_APP_PASSWORD)
        server.sendmail(Config.GMAIL_USER, email, msg.as_string())

def send_admin_notification(subject, body):
    msg = MIMEMultipart()
    msg['From'] = Config.GMAIL_USER
    msg['To'] = Config.ADMIN_EMAIL
    msg['Subject'] = subject
    msg.attach(MIMEText(body, 'plain'))
    with smtplib.SMTP('smtp.gmail.com', 587) as server:
        server.ehlo()
        server.starttls()
        server.login(Config.GMAIL_USER, Config.GMAIL_APP_PASSWORD)
        server.sendmail(Config.GMAIL_USER, Config.ADMIN_EMAIL, msg.as_string())

def send_admin_login_notification(user):
    login_time = datetime.now().strftime('%Y-%m-%d %H:%M:%S')
    ip = request.remote_addr
    body = f"""A user has just logged in to the Security System.
```

```
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```

```
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```

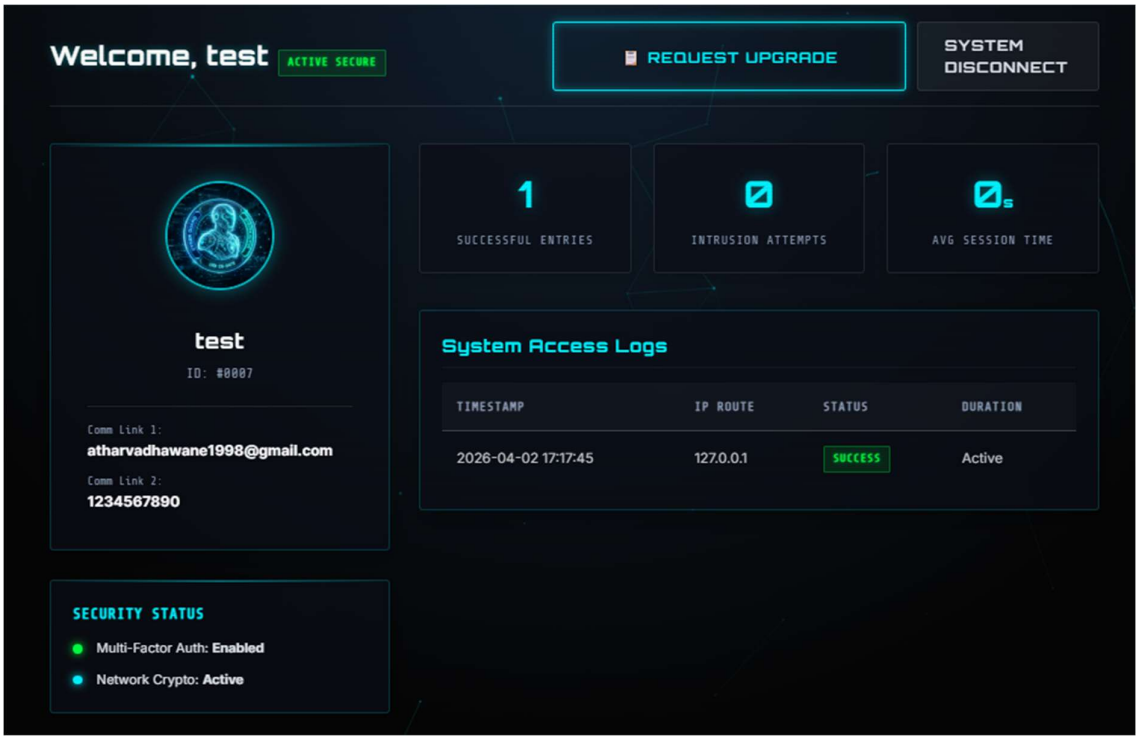
```
def log_login(user_id, status):
    ip = request.remote_addr
    db = get_db()
    try:
        with db.cursor() as cur:
            cur.execute("INSERT INTO login_logs (user_id, ip_address, status) VALUES
(%s,%s,%s)",
                        (user_id, ip, status))
        db.commit()
    finally:
        db.close()

def login_required(f):
    @wraps(f)
    def decorated(*args, **kwargs):
        if 'user_id' not in session:
            return redirect(url_for('login'))
        return f(*args, **kwargs)
    return decorated

def admin_required(f):
    @wraps(f)
    def decorated(*args, **kwargs):
        if 'admin_id' not in session:
            return redirect(url_for('admin_login'))
        return f(*args, **kwargs)
    return decorated
```

```
def get_user_by_identifier(identifier):
    db = get_db()
    with db.cursor() as cur:
        cur.execute("SELECT * FROM users WHERE email=%s OR mobile=%s", (identifier, identifier))
        user = cur.fetchone()
    db.close()
    return user
```

INPUT SCREEN & OUTPUT SCREEN



TESTING AND VALIDESTION CHECKS

TESTING AND VALIDATION CHECKS

Testing and validation are important phases in system development to ensure that the AI-Based Cyber Security Threat Detection System works correctly and meets user requirements. Testing helps identify errors, bugs, and system vulnerabilities, while validation ensures that the system fulfills security requirements and performs efficiently.

Testing

Testing is performed to verify that all modules of the AI-Based Cyber Security Threat Detection System function properly. Different types of testing are conducted during system development.

1. Unit Testing

Unit testing is performed on individual modules to check whether each module functions correctly. The modules tested include:

- User Registration Module
- Login Authentication Module
- OTP Verification Module
- Threat Detection Module
- User Activity Monitoring Module
- Admin Dashboard Module
- Account Lock Module
- Alert and Notification Module

2. Integration Testing

Integration testing is performed after combining different modules. This testing ensures that modules work together correctly. For example:

- Login system with OTP verification
- User activity monitoring with threat detection
- Failed login attempts with account lock system
- Admin dashboard with user monitoring system
- Alert system with admin panel

3. System Testing

System testing is performed to test the complete AI-Based Cyber Security Threat Detection System. This ensures that the entire system functions properly and securely.

This testing includes:

- System performance testing
- Security testing

• Functional testing
• Load testing

4. User Acceptance Testing

User acceptance testing is performed by administrators and users to verify that the system meets requirements. Users test features such as login system, OTP verification, and security alerts.

Validation Checks

Validation checks are applied to ensure correct data input and system security.

Input Validation

• Username validation
• Password validation
• Email format validation
• Mobile number validation
• OTP validation
• Required field validation

Data Validation

• Prevent empty fields
• Check valid input format
• Restrict invalid characters
• Password length validation
• OTP expiration validation

Security Validation

• Secure login authentication
• Password encryption
• Session management
• Account lock after multiple failed attempts
• Database security
• Admin authorization

Test Case Table

Test Case	Input	Expected Output	Result
User Login	Correct Username & Password	Login Successful	Pass
User Login	Wrong Password	Error Message	Pass

Test Case	Input	Expected Output	Result
OTP Verification	Correct OTP	Login Successful	Pass
OTP Verification	Wrong OTP	Error Message	Pass
Multiple Login Attempts	3 Wrong Attempts	Account Locked	Pass
Admin Unlock	Unlock User	Account Unlocked	Pass
User Registration	Valid Data	Registration Successful	Pass
Suspicious Activity	Multiple Attempts	Alert Generated	Pass

SECURITY MEASURES

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Security measures are implemented in the Portfolio Website System to protect user data, prevent unauthorized access, and ensure system reliability. The system includes various security features to safeguard sensitive information and maintain system integrity.

Authentication Security

The system provides a secure login mechanism for the administrator. Only authorized users can access the admin panel using valid username and password. Unauthorized users are restricted from accessing system features. This prevents misuse of system functionalities.

Password Security

Passwords are stored securely in the database. The system ensures that strong passwords are used for login authentication. Password protection prevents unauthorized users from accessing admin functionalities.

Session Management

Session management is implemented to maintain user login status. After successful login, a session is created and maintained until the user logs out. This helps prevent unauthorized access and improves system security.

Input Validation

Input validation is applied to all forms such as login form and contact form. This prevents invalid data entry and protects the system from malicious inputs. The system checks for empty fields, invalid characters, and incorrect formats.

Database Security

Database security measures are implemented to protect stored data. Only authorized users can access and modify database records. Database connections are secured to prevent unauthorized data access.

Access Control

Access control mechanisms are implemented to restrict user access. Only administrators have permission to modify system data. Visitors can only view information and submit contact forms.

Data Backup

Regular data backup is maintained to prevent data loss. In case of system failure, the data can be restored easily.

Error Handling

Error handling mechanisms are implemented to prevent system crashes and display user-friendly error messages. This improves system reliability and security.

These security measures ensure that the Portfolio Website System is secure, reliable, and protected from unauthorized access.

IMPLEMENTATION EVALUATION AND MAINTANANCE

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FUTURE SCOPE OF THE PROJECT

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The AI-Based Cyber Security Threat Detection System developed in this project provides an intelligent platform for detecting cyber threats and monitoring user activities. However, the system can be further enhanced by adding advanced features and technologies in the future. These improvements will increase system efficiency, scalability, and security.

In the future, the system can be enhanced by implementing advanced Machine Learning and Deep Learning algorithms for more accurate threat detection. This will allow the system to detect unknown and zero-day attacks more effectively. The system can also be improved by integrating real-time network traffic monitoring to detect threats at the network level.

Another future enhancement includes implementing behavioral analysis using Artificial Intelligence. This feature will analyze user behavior patterns and detect suspicious activities more accurately. The system can also be upgraded with predictive threat detection, which helps in identifying potential cyber threats before they occur.

The system can also be enhanced by adding multi-factor authentication (MFA) such as biometric authentication, face recognition, and fingerprint authentication. This will further improve system security and prevent unauthorized access.

Future improvements may also include:

- Cloud-based deployment for scalability
- Mobile application for remote monitoring
- Real-time notification system (Email & SMS alerts)
- Advanced admin dashboard with analytics
- Integration with firewall and IDS/IPS systems
- Automatic threat response system
- Blockchain-based security implementation
- Role-based access control system

SUGGETION & CONCLUSION

FUTURE SCOPE OF THE PROJECT

The AI-Based Cyber Security Threat Detection System developed in this project provides a smart and secure platform for detecting cyber threats and monitoring system activities. However, the system can be further enhanced by adding advanced features and technologies in the future. These improvements will increase the performance, scalability, and effectiveness of the system.

In the future, the system can be enhanced by implementing advanced Artificial Intelligence and Machine Learning algorithms to improve threat detection accuracy. This will help detect unknown cyber threats and zero-day attacks more efficiently. The system can also be upgraded to include real-time network traffic monitoring, which will help identify threats at an early stage.

Another future enhancement includes implementing user behavior analytics to monitor user activities and detect suspicious behavior. This will improve threat detection capabilities and reduce false alarms. The system can also be enhanced with automated threat response, where the system automatically blocks suspicious users and prevents potential attacks.

The system can also be upgraded by adding multi-factor authentication such as biometric authentication, fingerprint recognition, and facial recognition. These features will strengthen security and reduce unauthorized access risks.

Further improvements may include:

- Cloud-based deployment for better scalability
- Mobile application for remote monitoring
- Real-time alerts and notification system
- Advanced admin dashboard with analytics
- Integration with firewall and intrusion detection systems
- Role-based access control system
- Advanced reporting and logging system
- Automated security updates

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BIBLIOGRAPHY AND REFERENCES

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